# Digital Electronics Laboratory

# Laboratory Exercise 5. Counters

**Objectives:**

- To study the fundamentals of basic counters
- To construct various types of counters

**Introduction**

In this laboratory we will build on concepts that we examined in the previous laboratory sessions. We are now examining counters. We have not examined counters in detail at this stage, but please follow the steps outlined in the laboratory.

**Pre-Laboratory**

The task that you must perform **prior** to sitting this laboratory:

- Read the laboratory assignment in full.

**The Laboratory:**

**Section 1. The Ring Counter**

You may not have covered this material in the lectures as yet and you will be marked as appropriate to your knowledge. However, you will receive marks for initiative. Look at the circuits, see how they operate and try to work out why they would be useful. You are looking at counters today. Counters count. How do they count?

(a) Connect the JK flip flops as as shown in Figure 1.

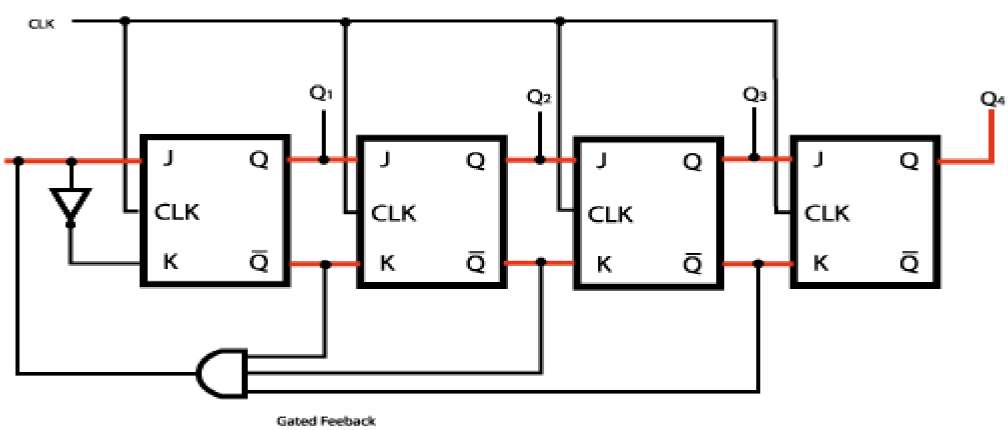

**Figure 1.** A Twisted ring counter

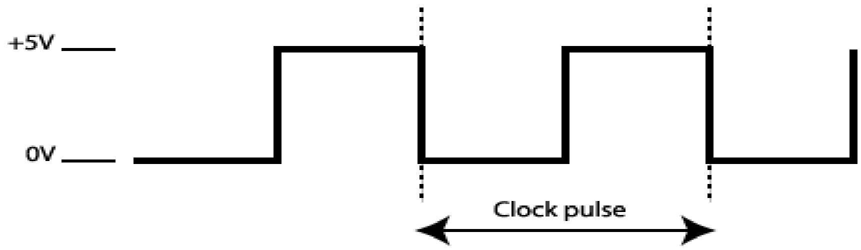


**Figure 2.** A typical clock signal.

(b) Apply clock pulses (as in Fig 2.) to the circuit and monitor the outputs of the counter (The Q values). What outputs do you get? (show in a table).

(c) In your opinion what is going on in this circuit? Is this circuit a counter? Why or why not?... How many different states can this circuit have when there are N flip flops?

**Section 2. The Asynchronous Counter**

(a) Connect the circuit for the asynchronous counter as shown in Figure 3. It consists of 3 JK flip-flops with the J and K inputs set to high.

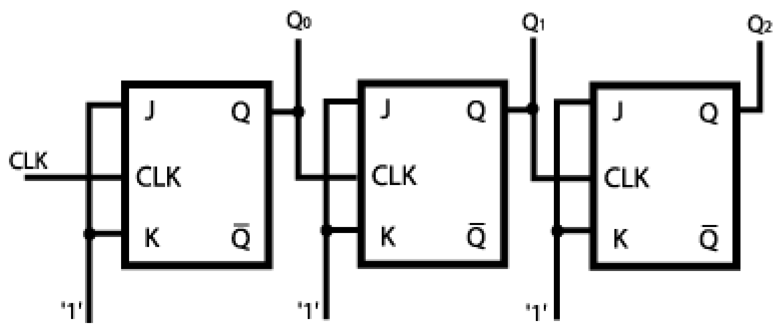


**Figure 3.** The Asynchronous Counter.

(b) Apply the clock pulses and measure the outputs at the Qn outputs. What values did you observe?

| | **Q0** | **Q1** | **Q2** |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |

| | | | |
|---|---|---|---|
| 9 | | | |
| etc.. | | | |

(c) How many states can this counter have for N flip-flops? In your opinion how does this circuit work? What could this circuit be useful for? Why do you think this is called an asynchronous counter, and not a synchronous counter (It has a clock?)?

**Section 3. Conclusions**

State briefly, but clearly, what you have gained from this laboratory.